UNTAN Programming Contest

Problem G

XOR Sum

Time limit: 1 second
Memory limit: 256 MB

Mr. Ucup has an array consisting of N integers. Mr. Ucup wants to know the sum of the XOR values of every pair of distinct elements.

Determine the value of:

$$\sum_{1 \le i < j \le N} (A_i \oplus A_j)$$

Operator $\oplus$ is the bitwise XOR operator.

Input

– The first line contains an integer N ($2 \le N \le 10^5$).

– The second line contains N integers $A_1, A_2, \ldots, A_N$ ($0 \le A_i < 10^9$).

Output

– Print the sum of the XOR values of every pair of elements.

Sample Input 1	Sample Output 1
4 1 2 3 4	24

Sample Input 2	Sample Output 2
3 5 5 5	0

Sample Input 3	Sample Output 3
5 0 1 2 4 8	60